

FinaBridge

Option D — Facilitator Model

AI Document Verification + Three-Tier Admin Review Pipeline + 13-Email Lifecycle

CHAIRMAN BRIEFING DOCUMENT

STRICTLY CONFIDENTIAL

PREPARED FOR

Reda — Director of Trade Finance

DIVISION

Finatrades Compliance & Technology

REFERENCE

FTD-BRIDGE-OPT-D-2025

VERSION

1.0 — Final Draft

DATE

22 March 2025

CLASSIFICATION

Strictly Confidential

~2 min

AI Verification

3 Tiers

Human Review

13

Email Touchpoints

7 Phases

Trade Lifecycle

100%

Audit Coverage

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01 Executive Summary

Strategic overview — FinaBridge Option D Facilitator Model

FinaBridge Option D transforms Finatrades from a passive marketplace into an active trade finance **facilitator**. Every trade request undergoes AI-powered document verification followed by structured three-tier human review before being published to exporters — delivering institutional-grade compliance, fraud prevention, and end-to-end audit coverage on every transaction.



AI Document Verification

OpenAI GPT-4o Vision reads Letter of Credit (LC), POL, and WR documents within minutes of submission. Extracts all critical fields — issuing bank, amount, expiry, beneficiary — and generates a fraud risk score from 0 (no risk) to 100 (critical). Documents scoring 50 or above are automatically rejected.



Three-Tier Human Review

Every AI-passed document then undergoes sequential human review: Macy (Operations Analyst, Tier 1) performs document verification; Farah (Compliance Officer, Tier 2) runs AML and sanctions screening; Director Reda makes the final publication decision. No trade reaches the marketplace without all three approvals.



Gold-Locked Settlement

Upon proposal acceptance, the importer's gold is locked in a cryptographic Settlement Hold — neutral custody held by Finatrades. The gold cannot be accessed by either party until the admin releases it after confirmed goods delivery. This guarantees payment to the exporter and protects the importer's assets.



Secure Deal Room

A real-time encrypted Deal Room is created between importer, exporter and admin upon proposal acceptance. Both parties must accept formal Terms & Conditions (DIAC arbitration, confidentiality, document authenticity warranty) before access. Full message history and all documents are archived permanently for regulatory purposes.

~2 min

AI Verification Speed

3 Layers

Human Review Depth

13+

Automated Email Alerts

7 Years

Document Retention

Business Objective

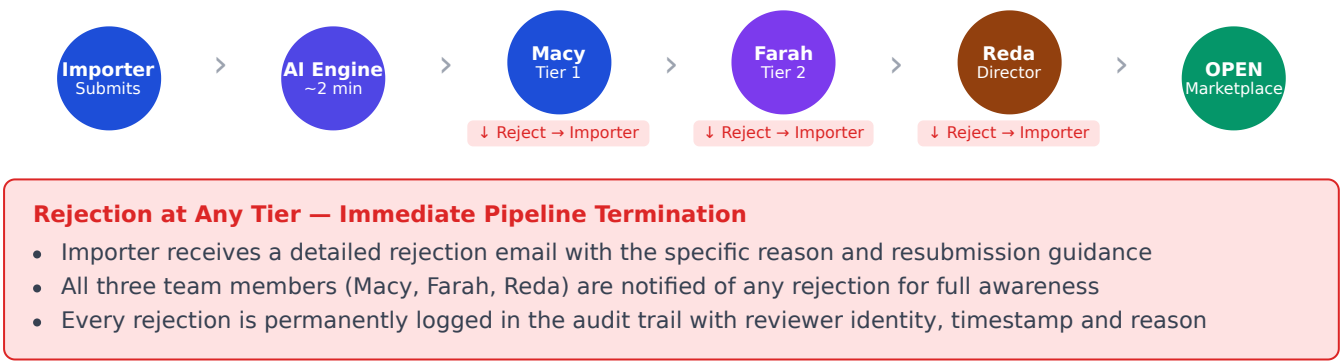
Position Finatrades as a trusted, compliant facilitator in international trade finance — generating platform fee revenue on every settled transaction while maintaining institutional-grade risk management that protects both importers and exporters.

02 Review Team & Responsibilities

Three-tier human review pipeline — accountability and escalation structure

TIER 1 Macy Operations Analyst - First-line document review - Verify AI extraction against original - Bank stamp & signature authenticity - Amount, date & beneficiary cross-check - Approve → Tier 2 or Reject with reason macy@finatrades.com Owns: Operations integrity — ensures AI output is accurate before compliance review proceeds.	TIER 2 Farah Compliance Officer - Full AML screening on both parties - Sanctions & PEP list check - Company registration verification - KYC cross-reference with documents - Approve → Director or Reject with reason farah@finatrades.com Owns: Regulatory compliance — ensures every trade meets AML, KYC and sanctions standards.	DIRECTOR Reda Director, Trade Finance - Reviews full audit trail (AI + Macy + Farah) - Makes final business & risk decision - Approves → publishes to marketplace - Rejects with Director-level reasoning - Full override authority at any tier reda@finatrades.com Owns: Final authority — business decision, risk appetite, and marketplace publication.
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Escalation Flow & Decision Points



03 Phase 1 — Trade Submission

Importer creates trade request: payment instrument selection, document upload, goods and shipping details

The Story — Ahmed Al-Rashid (Importer, Dubai)

Ahmed is a Dubai-based electronics importer. He logs into Finatrades and navigates to FinaBridge. As a first-time user, a **Terms & Conditions Disclaimer Modal** appears. He selects "Importer" as his role, digitally signs, and clicks Accept. He is now inside the FinaBridge dashboard and clicks **"Create Trade"**.

He completes the five-section submission form:

Section A — Payment Instrument (NEW)

Payment Instrument TypeNEW

Letter of Credit (LC) — Options: LC / POL / WR / FinaBridge Wallet

Document UploadNEW

LC_BankOfChina_2025.pdf (conditional — only shown for LC/POL/WR; max 10MB)

Section B — Goods & Trade Value

Goods Name

Electronic Components

Quantity

5,000 Units

Trade Value

USD 50,000.00

Gold Settlement

15.234 grams (auto-calculated from live gold price)

Payment Terms

30 days from Bill of Lading date

Section C — Shipping Details

Mode of Transport

Sea

Port of Loading

Shanghai, China

Port of Discharge

Jebel Ali, Dubai, UAE

Incoterms

FOB

Expected Ship Date

15 August 2025

Partial Shipment

Not Allowed

Transshipment

Not Allowed

Section D — Required Documents from Exporter

Required Documents

✓ Bill of Lading ✓ Commercial Invoice ✓ Packing List ✓ Certificate of Origin ✓ Insurance Certificate

Exporter Preference

No known exporter — Platform to suggest matching exporters

Key Change from Current System

The "Finance Type" field (previously a dead dropdown with only one useless option: "FinaBridge Finance") is completely replaced by **Payment Instrument Type** — giving importers the ability to select LC, POL, WR, or FinaBridge Wallet, with a conditional document upload zone that appears automatically for document-based instruments.

04 Phase 2 — AI Verification Engine

Automated document analysis, field extraction and fraud scoring within ~2 minutes of submission

Immediately upon submission, a background job is queued. The AI verification engine runs completely transparently — Ahmed sees "AI Review" status while the system processes his LC document:

1

Document Retrieved
LC PDF downloaded from Cloudflare R2 secure encrypted storage and prepared for analysis

2

Vision Analysis
OpenAI GPT-4o Vision processes the full document — reads all text, numbers, bank stamps, seals, signatures and tables simultaneously

3

Field Extraction
Structured data extracted: Issuing Bank, LC Number, Amount, Currency, Expiry Date, Beneficiary Name, Goods Description, Incoterms, Port details

4

Fraud Analysis — 6 Checks
Amount vs declared value | Expiry date validation | KYC beneficiary name match | Issuing bank registry verification | Document consistency | Format authenticity

5

Fraud Score Generated
Risk score 0–100 calculated. Score below 50 = PASS → Tier 1 Review. Score 50 or above = FAIL → AI Rejected. All flags documented.

6

Routing & Notifications
PASS: Status → "Tier 1 Review". Emails sent to Macy (action), Farah & Reda (awareness) and Ahmed (confirmation). FAIL: Auto-rejected, detailed emails sent immediately.

Sample AI Extraction — Ahmed's LC Document (PASSED)

✓ AI VERIFICATION PASSED		FRAUD SCORE: 8 / 100 — LOW RISK
Issuing Bank	Bank of China, Shanghai Branch	✓ Verified Institution
LC Number	LC2025-88442	✓ Valid Format
Amount	USD 50,000	✓ Matches Trade Value
Expiry Date	30 September 2025	✓ Future Date — Valid
Beneficiary	Ahmed Al-Rashid Trading LLC	✓ KYC Name Match
Goods	Electronic Components, 5,000 Units	✓ Consistent with Submission
Flags Raised	None	✓ Clean — No Anomalies

If AI FAILS — Alternate Path

- **Score ≥ 50** triggers immediate rejection with detailed flag report
- Example flags: Amount mismatch (LC: USD 30,000 vs declared USD 50,000) | Document expired (15 Jan 2025)
- Importer receives rejection email with specific reasons and resubmission guidance
- Macy, Farah and Reda all receive an awareness alert — no admin action needed

05 Phase 3 — Three-Tier Admin Review

Sequential approval: Macy (Tier 1) → Farah (Tier 2) → Director Reda — with mandatory notes and email trail

Macy — Tier 1: Operations Review

Macy receives an "ACTION REQUIRED" email. She opens Admin Panel → FinaBridge → Review Queue → Tier 1. Ahmed's trade shows AI fraud score (8/100) and all extracted fields. She opens the LC document in a side-by-side viewer with the AI report and verifies:

- ✓ Bank stamp is clear, authentic and matches the issuing bank in the AI extraction
- ✓ Amount of USD 50,000 visible on document matches trade declaration exactly
- ✓ Beneficiary name "Ahmed Al-Rashid Trading LLC" matches KYC on file
- ✓ LC expiry date (30 Sep 2025) is well within the trading window

DECISION: APPROVE → Tier 2 | Note: "Document verified. Bank stamp clear. Amount and beneficiary confirmed."

Email sent: Farah receives ACTION REQUIRED | Reda receives CC (awareness) | Macy receives confirmation copy

Farah — Tier 2: Compliance Review

Farah receives the Tier 1 approval email with Macy's notes. She opens Tier 2 queue and sees the complete review trail alongside the original document and AI report. She runs formal compliance checks:

- ✓ AML screening completed — no suspicious transaction patterns or structuring indicators
- ✓ Sanctions and PEP list check — neither Ahmed nor Wang Electronics appears on any watchlist
- ✓ Company registration confirmed against UAE trade registry database
- ✓ KYC documents fully cross-referenced — all details consistent and verified

DECISION: APPROVE → Director | Note: "AML clear. No sanctions match. Company registration confirmed against UAE trade registry."

Email sent: Reda receives FINAL DECISION REQUIRED | Farah + Macy receive CC (awareness)

Reda — Director: Final Approval

Reda receives the escalation email showing the complete review trail: AI passed (8/100), Macy approved with note, Farah approved with note. He opens the Final Approval panel with a single-screen view of all evidence — document, AI report, and both reviewer notes.

- ✓ Full audit trail complete: AI ✓ (8/100 LOW RISK) | Macy ✓ | Farah ✓
- ✓ Trade value (USD 50,000) within approved single-transaction risk limits
- ✓ LC instrument from verified issuing bank (Bank of China, Shanghai)
- ✓ No business, legal or risk concerns identified across any review layer

DECISION: APPROVE — Publish to Marketplace | publishedToExporters = true | Status → OPEN

Email sent: Ahmed receives "Trade is LIVE" email | Macy + Farah receive FYI completion copy

06 Phase 4 — Marketplace & Exporter Proposals

Trade published live to marketplace — exporter discovers, reviews and submits a structured proposal

Trade Goes Live — Marketplace Publication

The moment Reda approves, Ahmed's trade is instantly published. Status changes to **OPEN**. Ahmed receives a congratulatory email. The trade appears on the FinaBridge marketplace visible to all registered exporters:

TR-2025-0847 | Electronic Components | 5,000 Units | ● OPEN
Trade Value: USD 50,000 | Gold Settlement: 15.234g | Instrument: Letter of Credit
Route: Shanghai → Jebel Ali, Dubai | Incoterms: FOB | Mode: Sea Freight
Required Documents: B/L, Commercial Invoice, Packing List, Certificate of Origin, Insurance Certificate

Wang Electronics Co. Ltd — Exporter (Shanghai) Discovers & Submits

David Wang logs into Finatrades, switches to "Exporter" mode in FinaBridge, and opens the Marketplace. He filters by "Sea Freight" and "FOB" and finds Ahmed's trade. He clicks **"Submit Proposal"**:

Wang's Proposal — Submitted 23 March 2025	
Quote Price	USD 48,500 (USD 1,500 saving vs trade value)
Timeline	35 Days from order confirmation
Port of Loading	Shanghai Yangshan Deep-Water Port
Shipping Method	Sea — Full Container Load (FCL)
Incoterms	FOB
Estimated Delivery	28 August 2025
Insurance	Included in quote price
Certifications	ISO 9001:2015, CE Marking, RoHS Compliant
Company Name	Wang Electronics Co. Ltd
Registration No.	CN-SH-20180045
Contact Person	David Wang — Director of Export Operations
Contact Email	david@wangelectronics.cn
Notes	Exported similar components to GCC region 12 times. All certifications immediately available. ISO-certified facility.

Admin Forwarding (Optional Step)

The Finatrades admin reviews all incoming proposals and may shortlist / forward the best candidates to the importer's view. This acts as an additional quality filter before the importer sees proposals, ensuring only credible exporters reach the acceptance stage.

07 Phase 5 — Proposal Acceptance & Gold Lock

Importer accepts proposal — gold cryptographically reserved — Deal Room created simultaneously

Ahmed Reviews & Accepts Wang's Proposal

Ahmed receives email notification of a new proposal. He navigates to My Requests → TR-2025-0847 → View Proposals. He reviews Wang's offer against his requirements:

Criteria	Wang's Offer	Requirement	Assessment
Quote Price	USD 48,500	≤ USD 50,000	✓ USD 1,500 saving
Timeline	35 days	30–45 days	✓ Within range
Certifications	ISO 9001, CE, RoHS	Required	✓ All met
Insurance	Included	Required	✓ No extra cost
Port of Loading	Yangshan, Shanghai	Shanghai area	✓ Correct port

Ahmed Clicks "Accept Proposal" — Three Events Happen Simultaneously

1

Gold Reserved in Settlement Hold

15.234g of gold immediately locked from Ahmed's FinaBridge Wallet into a cryptographic Settlement Hold. Gold cannot be accessed, withdrawn or transferred by either party until released or cancelled by admin.

2

Deal Room Created

A secure Deal Room is instantiated with three participants: Ahmed (Importer), David Wang (Exporter), and an assigned Finatrades Admin. Real-time WebSocket connection established. Both receive access emails.

3

Status Updated to Active Trade

Trade status changes from "Proposal Review" to "Active Trade". All other pending proposals for this trade are automatically declined. Wang's proposal status changes to "Accepted".

Settlement Hold — Neutral Gold Custody

- The 15.234g is held in **Finatrades' custody** — not accessible to either party during the active trade
- Exporter is guaranteed payment exists and is secured before they ship any goods
- Importer's gold cannot leave until they confirm receipt of goods and admin authorises release
- Admin can cancel the hold (gold returns to importer) if trade fails or fraud is discovered

08 Phase 6 — Deal Room Operations

Secure real-time communication, document exchange and admin oversight — 4 to 6 weeks

Terms & Conditions — Both Parties Must Accept First

Both Ahmed and David Wang see a **T&C Modal** on first entry covering: Confidentiality, AML/KYC compliance, DIAC arbitration for disputes, gold settlement acknowledgement, document authenticity warranty, and UAE governing law. Both must tick "I agree" before accessing the chat.

Typical Deal Room Conversation (Aug 1 - Sep 15, 2025)

David Wang (Exporter) Shipment booking confirmed. Vessel: EVER GIVEN II. ETD Shanghai: 5 Aug 2025. Uploading Draft B/L for your review. Draft_BL_TR2025-0847.pdf	Aug 1
Ahmed Al-Rashid (Importer) Draft B/L received. One correction needed: consignee name should read "Ahmed Al-Rashid Trading LLC" — currently shows "Ahmed Trading LLC". Please correct and reissue.	Aug 2
David Wang (Exporter) Correction noted. Revised B/L issued by carrier. Revised_BL_FINAL_TR2025-0847.pdf Commercial_Invoice_\$48500.pdf PackingList_5000units.pdf	Aug 3
Ahmed Al-Rashid (Importer) Revised B/L confirmed ✓. Consignee name correct. Approved. Invoice and Packing List also reviewed and accepted.	Aug 4
David Wang (Exporter) Final documents uploaded. CertificateOfOrigin_China.pdf MarineInsurance_Certificate.pdf	Aug 6
Finatrades Admin (Official Notice) All 5 required documents received and verified: B/L ✓, Commercial Invoice ✓, Packing List ✓, Certificate of Origin ✓, Insurance Certificate ✓. Awaiting vessel arrival and importer delivery confirmation.	Aug 7
Ahmed Al-Rashid (Importer) Vessel arrived Jebel Ali Port. Goods inspected — 5,000 units of Electronic Components received in full. All items match packing list. Quality verified against specifications. Delivery confirmed ✓	Sep 12

Deal Room Features	
Real-time Messaging	Encrypted chat — visible to importer, exporter and assigned admin only
Document Exchange	All shipping documents uploaded, reviewed and stored with permanent access
Admin Oversight	Admin monitors all communications and can post official notices
Typing Indicators	Live presence — both parties see when the other is typing
Audit Log	Complete message history preserved permanently — even after closure

09 Phase 7 — Gold Settlement & Deal Room Closure

Admin processes gold release, trade completed, Deal Room permanently archived

Settlement Processing — Admin Action (15 Sep 2025)

Following Ahmed's delivery confirmation, the admin navigates to **FinaBridge Management → Settlement Holds** and locates TR-2025-0847. All conditions are verified:

- 1

Admin Verifies Conditions
All 5 documents uploaded and verified in Deal Room | Importer confirmed delivery in chat | No disputes raised within 48-hour review window
- 2

Admin Clicks "Release Settlement"
Single action triggers the full settlement cascade. The system processes all steps automatically with no manual transfers required.
- 3

Gold Debited from Settlement Hold
15.234g automatically removed from the cryptographic Settlement Hold (neutral Finatrades custody).
- 4

Platform Fee Deducted
e.g., 1% platform fee = 0.152g deducted automatically from settlement amount before transfer.
- 5

Gold Credited to Exporter
15.082g (net of platform fee) credited to David Wang's FinaBridge Wallet. Immediately available for conversion or withdrawal via FinaPay.
- 6

Trade Status → Completed
Trade request status updated to "Completed". Full transaction records created for both parties. Audit trail finalised.

Deal Room Closure — Admin Final Step

- ✓ Admin clicks "Close Deal Room" from inside the Deal Room interface
- ✓ Closure notes added: "Trade settled. All documents verified. Gold transferred 15 Sep 2025."
- ✓ Deal Room becomes permanently read-only — no new messages can be sent by any party
- ✓ Both Ahmed and Wang receive "Deal Room Closed" email with archive access link
- ✓ Full conversation history and all documents remain accessible for 7 years

Trade TR-2025-0847 — Complete Summary

Submitted: 22 Mar 2025 | AI Verified: 22 Mar 2025 | Approved: 22 Mar 2025 | Matched: 25 Mar 2025 | Active: 26 Mar 2025 | Completed: 15 Sep 2025

Duration: ~176 days | Gold Settled: 15.082g net | Platform Fee: 0.152g | Status: ✓ COMPLETED

10 Complete Email Notification Map

All 13+ automated email triggers — recipients, content summary and classification

Every email is triggered automatically by a platform event or status change. No manual sending is required. The system uses pre-built HTML templates with full Finatrades branding.

#1	✉	[FinaBridge] Trade Request Submitted — TR-2025-0847	USER EMAIL
TO: Ahmed (Importer) TRIGGER: Trade Submitted			
- Reference: TR-2025-0847 Goods: Electronic Components Value: USD 50,000 - Instrument: Letter of Credit (LC) Submitted: 22 March 2025, 10:42 AM GST - AI verification is in progress. You will be notified of the outcome within minutes.			
#2	✉	[FinaBridge Internal] New Submission — TR-2025-0847 AI Processing	INTERNAL
TO: Macy + Farah + Reda (simultaneously) TRIGGER: Trade Submitted			
- Importer: Ahmed Al-Rashid Trading LLC Value: USD 50,000 Instrument: LC - AI engine is running. No action required at this stage. - You will receive the verification result automatically within ~2 minutes.			
#3A	✉	[FinaBridge] 🟢 AI Passed — TR-2025-0847 ACTION REQUIRED: Macy	ACTION REQ
TO: Macy (ACTION) + Farah + Reda (CC) TRIGGER: AI Verification PASSED			
- Fraud Score: 8/100 LOW RISK All 6 checks passed No flags raised - Extracted: Bank of China LC2025-88442 USD 50,000 Exp: 30 Sep 2025 KYC Match ✓ - Macy: please complete Tier 1 review within 1 business day. [Open Admin Panel]			
#3B	✉	[FinaBridge] 🔴 AI FAILED — TR-2025-0847 Auto-Rejected	ALERT
TO: Macy + Farah + Reda + Ahmed (all) TRIGGER: AI Verification FAILED			
- Score: 78/100 HIGH RISK Flags: Amount mismatch (LC: \$30K vs declared \$50K) + Expired document - Request automatically rejected. Importer has been notified separately with detailed reasons. - No admin action required. Logged in audit trail for regulatory reporting.			
#4	✉	[FinaBridge] LC Verified — Under Admin Review	USER EMAIL
TO: Ahmed (Importer) TRIGGER: AI PASSED			
- Your Letter of Credit has passed automated compliance verification (Score: 8/100). - Your trade request is now progressing through internal admin review. - Estimated review time: 1-2 business days. You will be notified at each stage.			
#5	✉	[FinaBridge] Tier 1 Approved — TR-2025-0847 ACTION REQUIRED: Farah	ACTION REQ
TO: Farah (ACTION) CC: Reda + Macy TRIGGER: Macy Approves (Tier 1)			
- Reviewed by: Macy (Operations Analyst) Timestamp: 22 March 2025, 02:15 PM GST - Note: "Document verified. Bank stamp clear. Amount and beneficiary confirmed." - Farah: please complete Tier 2 compliance review within 1 business day. [Open Panel]			
#6	✉	[FinaBridge] Tier 2 Approved — TR-2025-0847 FINAL DECISION: Reda	ACTION REQ
TO: Reda (ACTION) CC: Farah + Macy TRIGGER: Farah Approves (Tier 2)			
- Trail: AI 🟢 (8/100) Macy 🟢 (22 Mar, 02:15) Farah 🟢 (22 Mar, 04:30) - Farah note: "AML clear. No sanctions match. Company registration confirmed." - Reda: your approval will publish this trade to all exporters. [Open Final Approval Panel]			

10 Email Notification Map (continued)

Emails 7-13: Trade approval, proposals, deal room, settlement and closure notifications

#7	✉ [FinaBridge] Trade APPROVED — TR-2025-0847 is Now Live	USER EMAIL
TO: Ahmed (Importer) TRIGGER: Reda Approves (Director)		
- Your trade has been reviewed and approved by our full compliance team. - TR-2025-0847 Electronic Components USD 50,000 Status: OPEN — Live on Marketplace - Registered exporters can now view and submit proposals. You will be notified as proposals arrive. [View Trade]		
#7B	✉ [FinaBridge Internal] Director Approved — TR-2025-0847 Published	INTERNAL
TO: Macy + Farah (FYI awareness) TRIGGER: Reda Approves		
- Director Reda has approved trade TR-2025-0847. Trade is now live on marketplace. - This is an awareness notification only — no action required from either of you.		
#8	✉ [FinaBridge] New Proposal Received — TR-2025-0847	USER EMAIL
TO: Ahmed (Importer) TRIGGER: Exporter Submits Proposal		
- Wang Electronics Co. Ltd (Shanghai) has submitted a proposal for your trade. - Quote: USD 48,500 Timeline: 35 Days Insurance: Included FOB Shanghai Delivery: 28 Aug 2025 [Review Proposal in FinaBridge Dashboard]		
#9	✉ [FinaBridge] Deal Room Open — 15.234g Gold Locked — TR-2025-0847	SUCCESS
TO: Ahmed (Importer) TRIGGER: Importer Accepts Proposal		
- You accepted Wang Electronics Co. Ltd's proposal for USD 48,500. 15.234g gold has been locked in Settlement Hold — your payment is secured. - A secure Deal Room is now open. Begin document coordination with your exporter. [Enter Deal Room]		
#10	✉ [FinaBridge] Your Proposal Was Accepted — Enter Deal Room	SUCCESS
TO: Wang (Exporter — David Wang) TRIGGER: Importer Accepts Proposal		
- Congratulations. Ahmed Al-Rashid has accepted your proposal for TR-2025-0847 (USD 48,500). - Gold settlement (15.234g) is cryptographically locked and ready for release upon confirmed delivery. - Please enter the Deal Room to begin shipment coordination and document exchange. [Enter Deal Room]		
#11	✉ [FinaBridge] Trade Completed — Gold Settlement Processed — TR-2025-0847	SUCCESS
TO: Ahmed (Importer) TRIGGER: Admin Releases Gold Settlement		
- Your FinaBridge trade TR-2025-0847 has been successfully completed and settled. - Gold Settlement: 15.234g transferred to exporter Platform Fee: 0.152g Completed: 15 Sep 2025 - Thank you for trading on FinaBridge. [Download Trade Certificate] [Start New Trade]		
#12	✉ [FinaBridge] Gold Settlement Received — TR-2025-0847	SUCCESS
TO: Wang (Exporter — David Wang) TRIGGER: Admin Releases Gold Settlement		
15.082g gold (net of platform fee) has been credited to your FinaBridge Wallet. - Trade reference TR-2025-0847 is now marked Complete. Full transaction record in your history. - You may convert or withdraw via FinaPay at any time. [Go to FinaPay] [View Transaction]		
#13	✉ [FinaBridge] Deal Room Closed — TR-2025-0847 Archive Available	CLOSURE
TO: Ahmed + David Wang (both parties) TRIGGER: Admin Closes Deal Room		
- The Deal Room for trade TR-2025-0847 has been formally closed by Finatrades admin team. - Closed by: Finatrades Admin Closed at: 15 September 2025, 10:30 AM GST - Closure note: "Trade settled. All documents verified. Gold transferred successfully." [View Archive]		

Quick Reference — All Emails at a Glance

#1	Trade Submitted	Ahmed (Importer)	USER EMAIL
#2	Trade Submitted	Macy + Farah + Reda	INTERNAL
#3A	AI Verification Passed	Macy (action) + Farah + Reda (CC)	ACTION REQ
#3B	AI Verification Failed	All 3 admins + Ahmed	ALERT
#4	AI Passed	Ahmed (Importer)	USER EMAIL
#5	Macy Approves	Farah (action) + Reda + Macy (CC)	ACTION REQ
#6	Farah Approves	Reda (action) + Farah + Macy (CC)	ACTION REQ
#7	Reda Approves	Ahmed (Importer)	USER EMAIL
#7B	Reda Approves	Macy + Farah (FYI)	INTERNAL
#8	Proposal Received	Ahmed (Importer)	USER EMAIL
#9	Proposal Accepted	Ahmed (Importer)	SUCCESS
#10	Proposal Accepted	Wang (Exporter)	SUCCESS
#11	Settlement Released	Ahmed (Importer)	SUCCESS
#12	Settlement Released	Wang (Exporter)	SUCCESS
#13	Deal Room Closed	Ahmed + Wang (both)	CLOSURE

11 Technology Architecture

AI stack, cloud infrastructure, storage, real-time communication and platform foundation

Layer	Technology	Purpose & Notes
● AI Vision Engine	OpenAI GPT-4o Vision	Multi-modal document reading — processes text, tables, bank stamps, seals, handwriting and signatures simultaneously from LC/POL/WR scans. Structured output via function calling.
● OCR Secondary	OCR.space Free API	500 free requests/day — fallback text extraction layer for documents where vision needs supplementary OCR confirmation. Zero cost.
● Job Queue	BullMQ + Redis (Upstash)	Async AI processing — document analysis runs in background without blocking the user session. Auto-retry on failure with exponential backoff. Priority queue for urgent reviews.
● File Storage	Cloudflare R2	Encrypted document storage — S3-compatible API. Direct URL access for AI analysis. Bucket: finatrades-gold. All uploads persist permanently with access logging.
● Database	PostgreSQL on AWS RDS	Singapore region (ap-southeast-1). Trade records, AI results, settlement holds, audit logs, deal room messages — all in one normalised schema. SSL enforced, automated backups.
● Real-time Chat	Socket.IO (WebSocket)	Deal Room messaging with live presence indicators, typing status and instant message delivery. Rooms scoped by deal room ID. Auto-reconnection on network loss.
● Email Service	Nodemailer	13 HTML email templates triggered by status changes. Full Finatrades branding. CC logic for internal awareness. No third-party ESP cost for current volume.
● Platform Stack	Node.js + Express + React 18	TypeScript throughout. Drizzle ORM. Tailwind v4 + shadcn/ui components. BullMQ for jobs. Session-based authentication with role-based permissions.

New Database Fields Required (Option D)

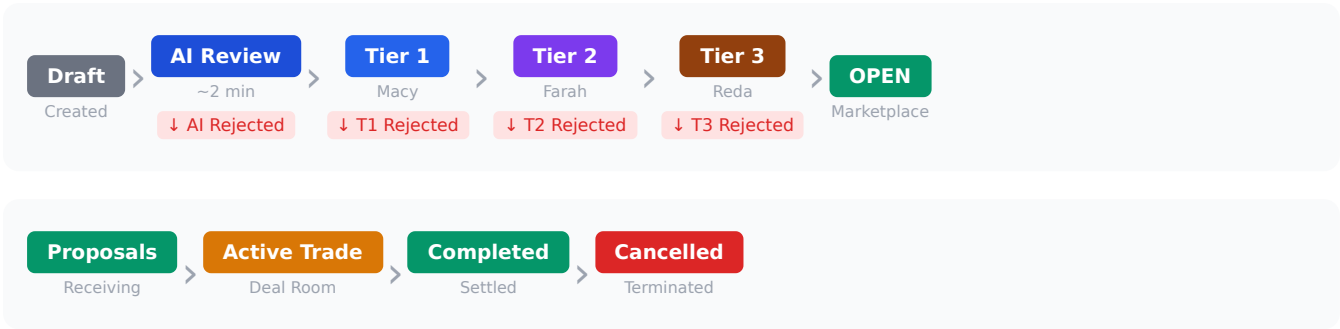
trade_requests table — New Columns

paymentInstrumentType	enum: LC POL WR Wallet	Payment instrument selected by importer
supportingDocumentUrl	text / url	Cloudflare R2 URL of uploaded LC/POL/WR document
aiVerificationStatus	enum: Pending Passed Failed	Result of AI verification run
aiFraudScore	integer 0-100	Fraud risk score from GPT-4o analysis
aiExtractedData	jsonb	All fields extracted from document by AI
aiRejectionReason	text	Human-readable reason for AI rejection
tier1/2/3Status	enum: Pending Approved Rejected	Individual decision at each review tier
tier1/2/3Notes	text	Reviewer notes at each tier
tier1/2/3ReviewedBy	uuid (userId)	Identity of reviewer at each tier
publishedToExporters	boolean	Set true by Reda approval — controls marketplace visibility

12 Status Flow Diagram

Complete trade lifecycle — all states, transitions, and rejection exit points

Every trade request moves through a defined set of statuses. Rejection at any AI or human review stage terminates the pipeline immediately and notifies all relevant parties.



Status Reference — Meaning & Responsible Party		
Draft	Trade created but not yet submitted by importer	Importer
AI Review	Document uploaded, AI engine processing in background	System (BullMQ)
AI Rejected	AI score ≥ 50 — document failed automated verification	System Auto
Tier 1 Review	Awaiting Macy's operations review	Macy
Tier 2 Review	Tier 1 passed — awaiting Farah's compliance review	Farah
Tier 3 Review	Tier 2 passed — awaiting Reda's final approval	Reda
OPEN	Published to marketplace — accepting exporter proposals	System
Proposal Review	Importer reviewing submitted exporter proposals	Importer
Active Trade	Proposal accepted, gold locked, Deal Room open	All Parties
Completed	Gold settled, trade closed successfully	System
Cancelled	Trade cancelled — gold returned to importer	Admin

13 Security, Compliance & Implementation Roadmap

Regulatory framework, legal structure and phased delivery plan with effort estimates

Compliance & Legal Framework

Anti-Money Laundering (AML)

- ✓ Farah (Tier 2) performs dedicated AML screening on every trade before Director review
- ✓ AI engine flags any structuring indicators (e.g., split amounts, unusual patterns) as high-risk
- ✓ All AML decisions logged with reviewer identity, timestamp and reasoning for regulatory audit

Know Your Customer (KYC)

- ✓ Importer KYC name automatically cross-matched against LC/POL/WR beneficiary field by AI
- ✓ Any name mismatch is flagged as high-risk fraud indicator — requires manual override to proceed
- ✓ Full verified KYC required before any user can access FinaBridge at all

Sanctions & Geo-Restrictions

- ✓ Farah performs sanctions list and PEP (Politically Exposed Person) screening on both parties
- ✓ Country-level geo-restrictions applied at platform level — blocked jurisdictions auto-reject
- ✓ No manual bypass possible without Director-level override with documented written justification

Dispute Resolution & Data Retention

- ✓ All Deal Room Terms reference Dubai International Arbitration Centre (DIAC) as exclusive forum
- ✓ Complete message history with timestamps preserved as legally admissible evidence permanently
- ✓ Document metadata (upload time, IP address, user identity) retained for 7 years per UAE commercial law

Implementation Roadmap — Four Phases

Phase 1 — Schema & Form Changes

3-5 days

Low Effort

- Add paymentInstrumentType field to trade_requests table
- Add supportingDocumentUrl, aiVerificationStatus, aiFraudScore, aiExtractedData, aiRejectionReason fields
- Add tier1/2/3 status, notes and reviewedBy fields to schema
- Extend status enum: AI Review, AI Rejected, Tier 1/2/3 Review values
- Update FinaBridge form UI with Payment Instrument selector and conditional upload zone

Phase 2 — AI OCR Verification Service

5-7 days

Medium Effort

- Build server/services/ocr-service.ts — central AI document analysis engine
- BullMQ job queue for asynchronous, non-blocking, retry-safe document processing
- GPT-4o Vision integration with document-type-specific prompts for LC, POL and WR
- Fraud scoring engine implementing all 6 check points (amount, expiry, KYC, bank, consistency, format)
- Admin AI report viewer: side-by-side original document and extracted data display panel

Phase 3 — Three-Tier Admin Review Panel

5-7 days

Medium Effort

- Add Tier 1, Tier 2, and Final Approval queue tabs to FinaBridgeManagement.tsx
- Document viewer with AI extraction report in side-by-side split panel layout
- Approve and Reject action buttons with mandatory review notes at each tier
- Role-based access control: Macy sees Tier 1 only, Farah sees Tier 2 only, Reda sees all
- Full review trail display on every trade card (AI score + Macy notes + Farah notes)

Phase 4 — Email Notification System (13 Templates)

📅 3-4 days

Low Effort

- 13 professional HTML email templates with complete Finatrades branding and purple/gold colour scheme
- Trigger hooks implemented at every status change point throughout routes.ts
- Admin team emails: Macy, Farah, Reda — simultaneous on submission, AI result and each tier
- CC logic: Macy CC'd on Tier 2 approval; Farah and Macy CC'd on Director approval
- Importer and Exporter emails at all user-facing events: submission, verification, approval, deal room, settlement



Ready for Chairman Approval

This document provides the complete technical, operational and compliance blueprint for FinaBridge Option D — Facilitator Model.

- ✓ Full AI document verification pipeline with 6-point fraud scoring engine
- ✓ Structured 3-tier human review: Macy → Farah → Director Reda
- ✓ 13 automated email notifications covering every touchpoint
- ✓ Gold-locked Settlement Hold — cryptographic neutral custody
- ✓ Secure real-time Deal Room with permanent audit trail
- ✓ Full regulatory compliance: AML, KYC, Sanctions, DIAC, UAE Law
- ✓ Implementation: 4 phases — estimated 16-23 development days total

Finatrades Technology & Compliance Division

Reference: FTD-BRIDGE-OPT-D-2025 | Version 1.0 | 22 March 2025 | STRICTLY CONFIDENTIAL